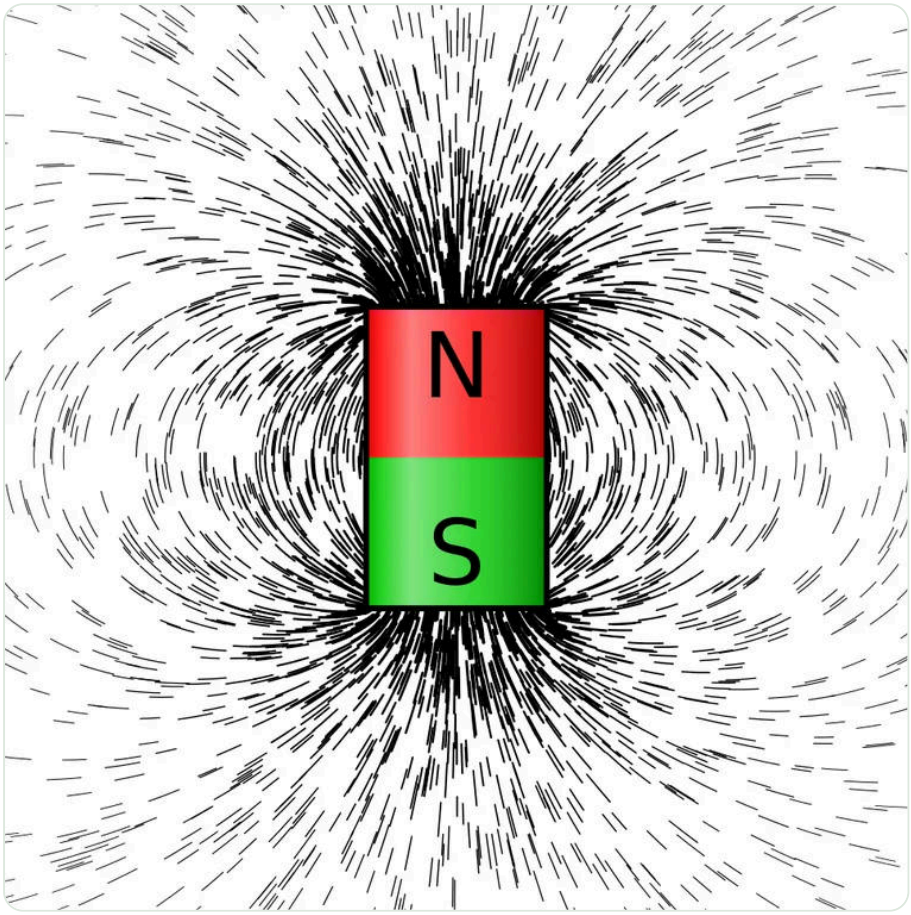
Sindh Science Learning Portal

Class 5 · Science

Chapter 8

Electricity and Magnets



Learn about uses of electricity and safety, a simple circuit, conductors and insulators, and the poles of a magnet.

Following the Sindh Textbook Board curriculum

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CHAPTER 8

Electricity and Magnets

What we will learn Learn about uses of electricity and safety, a simple circuit, conductors and insulators, and the poles of a magnet.

1 Electricity and safety

Electricity is a kind of energy. It lights bulbs, runs fans, cools the fridge and charges mobile phones.

We get electricity from **power stations** through wires. Small things like torches and clocks run on a **battery (cell)**.

Electricity is useful but it can also be **dangerous**:

- Never touch a switch or socket with wet hands.
- Never push anything into a socket.
- Do not fly kites near electric wires and poles.
- If you see a broken wire, do not touch it — tell an adult.

2 A simple circuit

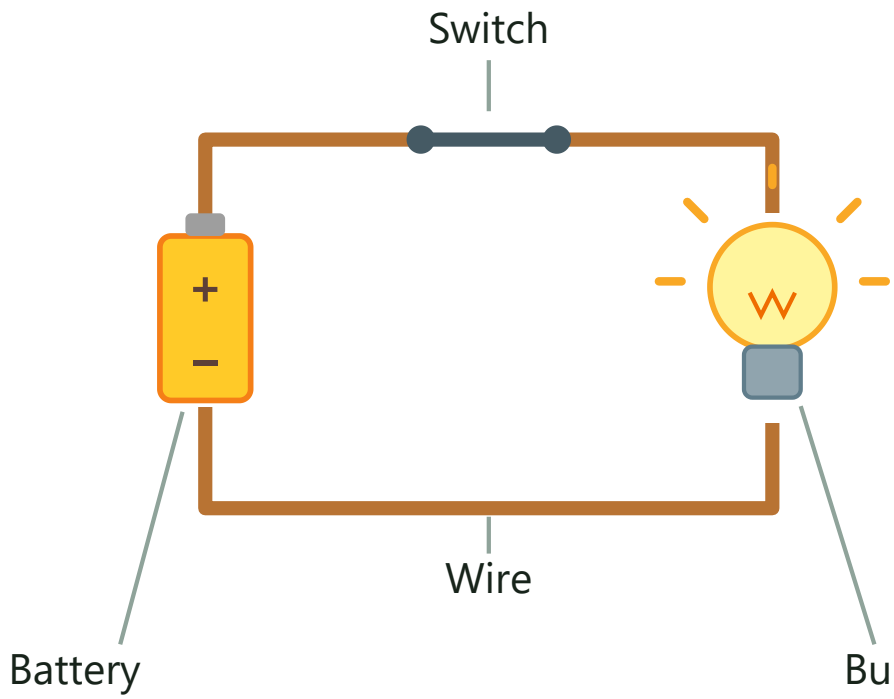
The path that electricity flows along is called a **circuit**. A simple circuit has:

- a **battery**, which pushes the electricity,
- **wires**, which carry the electricity from one place to another,
- a **switch**, which opens or closes the circuit,
- a **bulb**, which changes electricity into light.

When the path is complete (a **closed circuit**), the bulb lights up. When the path is broken somewhere (an **open circuit**), the bulb does not light.

Materials that let electricity pass are called **conductors**, like copper, iron and aluminium. Materials that do not let it pass are called **insulators**, like plastic, rubber, wood and glass. That is why wires are covered in plastic.

Diagram: the parts and what they do



Battery (بيتري · بیٹری)
Gives electrical energy and pushes electricity around the circuit.

Wire (تار · تار)
Copper wire carries electricity. Copper is a good conductor.

Switch (سوئچ · سوئچ)
Opens or closes the circuit. When closed, electricity flows and the bulb lights.

Bulb (بلب · بلب)
Changes electricity into light. It glows only in a complete circuit.

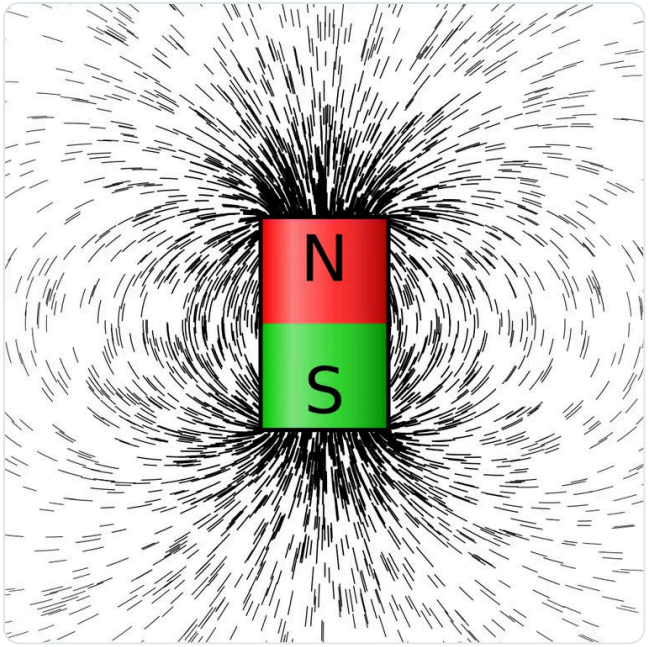
3 Magnets

A **magnet** pulls things made of **iron, steel and nickel** towards it. It does not pull paper, plastic or wood.

Every magnet has two **poles**: a **north pole (N)** and a **south pole (S)**. The pull is strongest at the ends of the magnet.

- **Unlike poles** (N and S) **attract** each other.
- **Like poles** (N and N, or S and S) **push each other away** (repel).

A magnet that hangs freely always points north–south. A **compass** works this way and helps us find our way.

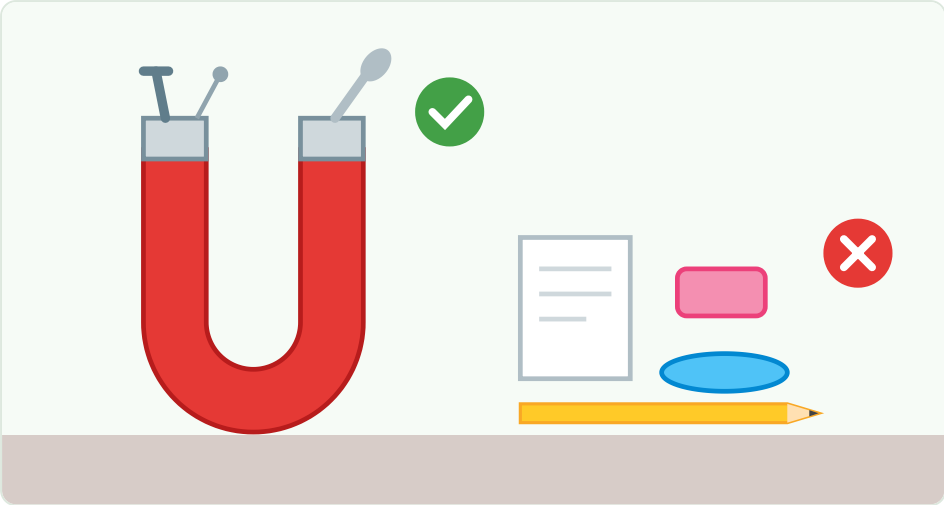


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Key words

English	سنڌي	اردو
electricity	بجلي	بجلی
battery (cell)	بيٽري (سيل)	بیٹری (سیل)
circuit	سرڪٽ	سرکٹ
switch	سوئچ	سوئچ
conductor	موصل	موصل
insulator	غير موصل	غیر موصل
magnet	مقناطيس	مقناطیس
pole	قطب	قطب
attract	چڪڻ	کشش
repel	پري ڌڪڻ	دفع کرنا
compass	قطب نما	قطب نما

Experiment: What does a magnet pull?



You need

- A magnet (a fridge magnet will do)
- An iron nail, a pin, a steel spoon
- Paper, a plastic lid, an eraser, a wooden pencil

Steps

1. Put all the things on a table.
2. Guess which things will stick to the magnet and write your guesses in your notebook.
3. Bring the magnet close to each thing.
4. Make two lists: pulled and not pulled. Compare them with your guesses.

Safety: Keep magnets away from mobile phones, watches and computers. Handle the pin carefully.

What did we learn? The nail, pin and steel spoon stuck to the magnet because they are made of iron or steel. Paper, plastic, rubber and wood were not pulled. A magnet pulls only some metals.

Practice questions

Circle the correct answer

1. **What makes a torch work?**

   (a) A battery
   (b) Water
   (c) A magnet
   (d) Paper

2. **Which of these is safe?**

   (a) Pressing a switch with wet hands
   (b) Pushing a pin into a socket
   (c) Flying a kite in an open field away from wires
   (d) Touching a broken wire

3. **When does the bulb in a circuit light up?**

(a) When the circuit is complete (closed)
(b) When a wire is broken
(c) When there is no battery
(d) When the switch is open

4. **Which of these is a conductor?**

(a) Plastic
(b) Rubber
(c) Copper
(d) Wood

5. **Why are electric wires covered in plastic?**

(a) Plastic is an insulator and keeps us safe
(b) To make the wire look pretty
(c) To make electricity faster
(d) To make the wire heavy

6. **What does a magnet pull?**

(a) An iron nail
(b) Paper
(c) A plastic spoon
(d) A wooden pencil

7. **What happens when the N poles of two magnets are brought close?**

(a) They push each other away
(b) They stick together
(c) They melt
(d) Nothing at all

8. **Which tool uses a magnet to help us find directions?**

(a) A compass
(b) A thermometer
(c) A weighing scale
(d) A watch

9. **What does a switch do in a circuit?**

(a) Opens or closes the circuit
(b) Makes electricity
(c) Gives light
(d) Makes a magnet

Write true or false

10. **Rubber is an insulator.** _____

11. **A bulb lights up in an open circuit.** _____

12. **A magnet has two poles.** _____

13. **A magnet pulls paper.** _____

14. **Touching a socket with wet hands is dangerous.** _____

Match the columns

15. **Match each part of a circuit with its job.**

1. Battery		(a) Carries the electricity
2. Wire		(b) Opens or closes the circuit
3. Switch		(c) Gives light

4. Bulb		(d) Pushes the electricity

Match the columns

16. Match each pair of poles with what happens.

1. N and S		(a) Push each other away
2. S and S		(b) Attract each other

Worksheet: Electricity and Magnets

Name: _____ Date: _____

A. Fill in the blanks

Word box: battery · compass · conductors · poles · switch

1. A torch runs on a _____ .
2. Materials that let electricity pass are called _____ .
3. A _____ opens or closes a circuit.
4. A magnet has two _____ .
5. A _____ helps us find directions.

B. Put ✓ for true and ✗ for false

1. ☐ Electricity passes easily through plastic.
2. ☐ A bulb lights up in a closed circuit.
3. ☐ Unlike poles attract each other.
4. ☐ It is safe to fly a kite near electric wires.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Copper	( ) Gives light
2. Rubber	( ) Push away
3. Bulb	( ) Conductor
4. N and N poles	( ) Insulator

D. Answer in short

1. Write two ways to stay safe with electricity.

_____

_____

2. Name the four parts of a simple circuit.

_____

_____

3. Write two uses of magnets.

_____

_____

E. Draw and label

Draw a simple circuit with a battery, wires, a switch and a bulb, and label each part.

FOR TEACHERS

Answer key

Chapter 8: Electricity and Magnets

Practice questions

1. (a) A battery
2. (c) Flying a kite in an open field away from wires
3. (a) When the circuit is complete (closed)
4. (c) Copper
5. (a) Plastic is an insulator and keeps us safe
6. (a) An iron nail
7. (a) They push each other away
8. (a) A compass
9. (a) Opens or closes the circuit
10. True
11. False
12. True
13. False
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. battery · 2. conductors · 3. switch · 4. poles · 5. compass

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Gives light · (4) Push away · (1) Conductor · (2) Insulator

D. Answer in short:

1. Do not touch switches with wet hands and never push anything into a socket.
2. Battery, wires, switch and bulb.
3. In a compass, and to keep a fridge door shut.

Picture credits

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